Example 9.17

Calculate $\lim_{t\to 1}\frac{\sqrt{t}-1}{t-1}$.

Solution

$$\lim_{x\to 1}\frac{\sqrt{t}-1}{t-1}=\lim_{x\to 1}\frac{t^{\frac{1}{2}}-1^{\frac{1}{2}}}{t-1}=\frac{1}{2}(1)^{\frac{1}{2}-1}=\frac{1}{2}.$$

Example 9.18

Find $\lim_{x\to 0}\frac{(2+x)^5-2^5}{x}$.

Solution

Put $2+x=y$ so that as $y\to 2$ as $x\to 0$.

Therefore, $\lim_{x\to 0}\frac{(2+x)^5-2^5}{x}=\lim_{y\to 2}\frac{y^5-2^5}{y-2}=5(2^4)=80$.

Example 9.19

Find the positive integer n so that $\lim_{x\to 3}\frac{x^n-3^n}{x-3}=27$.

Solution

$\lim_{x\to 3}\frac{x^n-3^n}{x-3}=n.3^{n-1}=27$

That is $n.3^{n-1}=3\times 3^2=3\times 3^{3-1}\Rightarrow n=3$.

Example 9.20

Find the relation between a and b if $\lim_{x\to 3} f(x)$ exists where $f(x)=\begin{cases} ax+b & \text{if } x>3 \\ 3ax-4b+1 & \text{if } x<3 \end{cases}$.

Solution

$\lim_{x\to 3^-} f(x)=9a-4b+1$

$\lim_{x\to 3^+} f(x)=3a+b$. Now the existence of limit forces us to have

$\lim_{x\to 3^-} f(x)=\lim_{x\to 3^+} f(x)$.

$\Rightarrow 9a-4b+1=3a+b$

$\Rightarrow 6a-5b+1=0.$

EXERCISE 9.2

Evaluate the following limits :

(1) $\lim_{x\to 2}\frac{x^4-16}{x-2}$

(2) $\lim_{x\to 1}\frac{x^m-1}{x^n-1}$, m and n are integers.

(3) $\lim_{\sqrt{x}\to 3}\frac{x^2-81}{\sqrt{x}-3}$

(4) $\lim_{h\to 0}\frac{\sqrt{x+h}-\sqrt{x}}{h}, x>0$

(5) $\lim_{x\to 5}\frac{\sqrt{x+4}-3}{x-5}$

09_11th-Maths-Vol2-Unit9.indd 102 16-12-2022 11:03:16